

ANIMESH RAJ

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EDUCATION

Indian Institute of Technology, Kharagpur

B.Tech (Hons.) in Electrical Engineering (Instrumentation Engineering) | CGPA: 7.73/10

Kharagpur, India

Aug 2023 – Apr 2027

Minor: Artificial Intelligence & Computer Science Micro-Specialization: High-Performance Computing

Delhi Model Public School

Class XII AISSCE (Science Stream) | Aggregate: 94.2%

Patna, India

Completed 2023

EXPERIENCE

VoiceEval

Remote

Software Engineer Intern (R&D) | **LangGraph**, **Vapi**, **LiveKit**, **Pipecat**, **Bolna**

Oct 2025 – Nov 2025

- Shipped QA platform at **voiceeval.com** unifying **6** voice AI frameworks (Vapi, LiveKit, LangGraph, Pipecat, Bolna, LangChain).
- Devised graph-based test generator converting unstructured prompts into automated suites; delivered **4x** faster voice-agent QA.
- Instrumented latency, tone, and logic-flow metrics across **over 1,000** runs; hardened CI stress-testing for generative voice models.

Convosight Analytics

Remote

Software Engineering Intern | **React**, **Supabase**, **NLP**

Aug 2025 – Oct 2025

- Migrated legacy workflows to a high-performance React/Supabase architecture, cutting infra costs by **38%** and load times by **50%**.
- Built ETL pipeline processing **over 10,000 daily posts** with **92%** sentiment accuracy via NLP for cross-market trend detection.
- Streamlined executive insight summarization via fine-tuned GPT systems, reducing manual content creation overhead by over **70%**.

Aristhrottle

Remote

AI/ML Engineer | **LangGraph**, **PostgreSQL**, **FastAPI**

Jun 2024 – Jul 2024

- Launched production story generation pipeline wiring Google Imagen3 for frame-level visual coherence and PostgreSQL for state.
- Designed stochastic prompt selector with temperature scaling, increasing creative output diversity by **25%** in A/B tests.
- Integrated **LangFuse SDK** to trace LangGraph's Stategraph executions, enabling granular monitoring of token usage and latency.

PROJECTS

AgentGuard: Multi-Agent Security & Governance Platform

Jan 2026 – Apr 2026

2nd Runner Up, Microsoft AI Unlocked 2026 | **Azure AI**, **River**, **NetworkX**, **OpenTelemetry**

- Architected 4-layer shield on OWASP Agentic Top 10: input regex and Azure Content Safety, output guards, tool firewall, anomaly.
- Ran **MELON** prompt-injection detection on tool calls; wired Azure Text Analytics and LLM-as-judge groundedness on outputs.
- Validated across **218 unit tests** and 5 agent-type pairs; shipped live **OpenTelemetry** audit dashboard at agentguard.exempl4r.xyz.
- Authored PBAC (**12 roles**) and **River HalfSpaceTrees** anomaly baseline; SQLite audit log with HITL/AITL approval gates.

TriageGym: RL Environment for AI Incident Triage

Mar 2026 – Apr 2026

Grand Finalist, Meta x PyTorch OpenEnv Hackathon: Top 800/31,000 Teams | **Python**, **Gymnasium**

- Authored RL benchmark with **12 fault patterns** (OOM, DNS, deadlocks, partitions), **7** investigation actions, 3 difficulty tiers.
- Modeled **sigmoid-curve** temporal degradation over NetworkX DAGs; **183 tests** guard generator, grader, and API determinism.
- Benchmarked **9 frontier models** scoring **0.52–0.97** via evidence-grounded grading that blocks reward-hacked diagnoses.

AgroWise: Phygital Precision Agriculture Platform

Jan 2026 – Mar 2026

1st Prize NSCIF 2026 | Hacksagon 2026 Finalist | **Embedded C**, **FastAPI**, **LLM**, **IVR**

- Designed AgroSense, a Rs 800 solar sensor transmitting root-zone soil data over cellular networks with zero internet dependency.
- Launched **Kisan-Vani** IVR delivering AI crop insights in Hindi/Bengali/Odia via phone for India's 14 crore smallholder farmers.
- Trained fusion model on leaf imagery and soil NPK to split fungal stress from nutrient deficiency, preventing Rs 500 per misdiagnosis.
- Won **Rs 50,000 cash prize** at NSCIF 2026; shortlisted for AMD Slingshot 2026 demo day and Hacksagon 2026 national finals.

OvertureAI: Automated Cold Outreach Platform | React 19, FastAPI, LangGraph, Supabase, GCP Cloud Run

Mar 2026

- Architected **7-node** multi-step pipeline from lead discovery through Gemini email, Telegram approval gate, to 14-day sequences.
- Incorporated DuckDuckGo, Gmail OAuth2, Telegram, Supabase Realtime, Vertex AI on **GCP Cloud Run** with a **59-test suite**.
- Engineered IIT KGP alumni discovery module; domain-affinity scoring cuts irrelevant leads by **60%** before Gemini email generation.

AlphaStream India: Real-Time Investment Intelligence Platform

Jan 2026 – Apr 2026

Finalist, ET GenAI Hackathon 2026 (Top 20 Nationally) | **Pathway**, **Gemini**, **DuckDB**

- Launched Bloomberg Terminal-style platform via **Pathway** with sub-2-second Nifty 50 latency for India's 15 crore investors.
- Orchestrated **13** specialised agents across portfolio analysis, Text2SQL NLQ, and anomaly detection via **River HalfSpaceTrees**.
- Merged **5** news feeds (NewsAPI, Finnhub, Alpha Vantage, MediaStack, RSS) and SEC EDGAR filings into DuckDB via WebSocket.

Aksha Drishti: Agentic Vision-Language System for Remote Sensing Imagery

Nov 2025 – Dec 2025

Gold Medal Winner, ISRO Inter IIT Tech Meet 14.0 | vLLM, Qwen3-VL, SAM3 |  Link

- Achieved **2-3x** latency reduction and **95%** GPU utilization via vLLM service optimized with KV caching and concurrent batching.
- Outperformed **GPT-4o** on VRSBench via DPO-aligned **Qwen3-VL-8B** pipeline; reached **87%** VQA accuracy on satellite imagery.
- Achieved **mIoU** of **0.73** in visual grounding via FastAPI **SAM3 Agent** with pyramidal tiling, cross-verified by VLM logic layer.

instiGPT: Universal Faculty Profile Extractor | GPT-4o Vision, Playwright, SQLAlchemy, DuckDuckGo

Nov 2025

- Built autonomous faculty scraper via **GPT-4o vision** to parse React/Angular/DataTables sites that defeat regex tools.
- Structured output into emails, Scholar h-index, and research areas; batch mode processes any university department URL to CSV.
- Parallelized Playwright sessions for multi-department crawls; **SQLAlchemy** deduplication prevents re-scraping on incremental runs.

RESEARCH & PUBLICATIONS

RAD-Agent: Voice-Driven Dataset Automation | Microsoft Presidio, Whisper

Mar 2025 – Jan 2026

- Composed **5-stage** pipeline (WhisperX, Presidio, RadGraph, FAISS, llama3-med42-8b) cutting annotation cost by **70–80%**.
- Enforced PHI de-identification via Microsoft Presidio and SymSpell correction; WhisperX held **WER** < **18%** on clinical audio.
- Achieved **F-score** **85–90%** and BERTScore **85–90%** on 5 radiology sets; medllama2:7b validator gates low-confidence fields.

CPRNet: Few-Shot Motor Fault Diagnosis

Jan 2026 – Present

Targeting IEEE Transactions on Industrial Informatics | PyTorch, ResNet18, GRU

- Devised **CPRNet** (ResNet18, soft attention, GRU T=3) hitting **98.78%** 5-shot accuracy on **27-class** motor fault diagnosis.
- Surpassed ProtoNet by **3.48pp** and MAML by **44.96pp** on 54K samples; CWRU cross-modal holds **95.30%** without retuning.
- Ablated Phase-2 meta-training to confirm load-bearing role (**-37.33pp**); deployed on Jetson Orin NX at **1.82 ms** inference.

COMPETITIONS

AMD Slingshot 2026: Regional Demo Day Finalist | Sustainable AI & Green Tech | AgroWise | Kolkata

Apr 2026

- Shortlisted nationally for Regional Demo Day in Kolkata; presented live AgroWise prototype to AMD technical jury on Apr 24–25.
- Demonstrated AgroWise under Sustainable AI theme; stack spans embedded C sensors, edge LLM, and rural IVR delivery.
- Optimized edge stack to AMD power targets; design constrained to sub-watt solar budget with offline cellular delivery.

MediaFlowAI: Video Publishing Analytics | GC 2026 Data Analytics | Frammer AI | FastAPI, DuckDB, LangGraph

Mar 2026

- Tracked **19 KPIs** across **4,569 video events** in 5 workspaces, surfacing PCR variance (38–92%) and pipeline bottlenecks.
- Incorporated **Gemini models** for insight generation and **Chronos-Bolt-Tiny** time-series forecasting; validated via **134 test cases**.
- Modeled DuckDB columnar schema enabling sub-100ms analytical queries on over 4,000 event logs without dedicated OLAP cluster.

VERITAS: Hallucination-Free RAG Pipeline | Winner, KDSH 6.0 | Over 13,500 Registrations | Pathway-Sponsored

Jan 2026

- Secured **1st Place** with a **5-layer constraint pipeline** verifying **100k-word** narratives end-to-end via **LangGraph** orchestration.
- Devised Evidence Ledger tracking missing facts via **FactScore** extraction, achieving **91% precision** and **83% accuracy**.
- Integrated **HippoRAG** graphs and **RAPTOR** tree retrieval via **Pathway** indexing; structurally eliminated hallucinations.
- Launched on **Modal** serverless GPU with autoscaled inference; end-to-end KDSH eval latency under **4s** per 100k-word narrative.

Netflix Strategic Intelligence Dashboard | Winner, Open IIT Data Analytics 2025 | Think Again Lab

Nov 2025

- Delivered interactive **React/Vite** dashboard mapping content strategies across **over 8,800** Netflix titles for cross-regional analysis.
- Trained NLP sentiment engine achieving **92% accuracy** and surfaced **10** actionable global market insights through reporting.
- Clustered titles by genre saturation; flagged **6** content gaps and **3** high-growth regional markets for commissioning strategy.

Employee Selection Engine | Open Soft GC 2025 | Bronze Winning Team

Apr 2025

- Architected predictive analytics pipeline achieving **87%** accuracy in at-risk identification across **6** heterogeneous HR datasets.
- Applied SMOTE to balance attrition labels; benchmarked **6** classifiers and calibrated XGBoost for recall-optimized risk flagging.
- Derived per-employee SHAP risk rankings; deployed Flask explainability dashboard with live attrition probability scoring.

Budget Optimization | Data Analytics GC 2025 | Bronze Winning Team

Mar 2025

- Modeled Marketing Mix with Meta's Robyn library to optimize an **\$18.7M** yearly budget, projecting **31.9%** revenue uplift.
- Calibrated per-channel adstock-decay and Hill-saturation curves; flagged **3** underperforming offline channels for budget reallocation.
- Generated Pareto budget frontiers with response-curve plots; presented allocation deck to **5-judge** GC panel earning Bronze.

TECHNICAL SKILLS

Languages & Core: Python, C++, TypeScript/JavaScript, SQL (PostgreSQL), Embedded C

AI & Agentic Systems: LangGraph, PyTorch, vLLM, Pathway, HuggingFace, OpenCV, SAM3, Whisper, Gymnasium

Web & Backend: React 19, FastAPI, Node.js, Supabase, DuckDB, Pinecone, FAISS, MongoDB, WebRTC

Cloud & Infra: Azure AI (Content Safety, Prompt Shields), GCP (Vertex AI), AWS, Modal, Docker, OpenTelemetry, Git/Linux

RELEVANT COURSEWORK

EE / Instrumentation: Signals & Systems, Control Systems, Digital Signal Processing (DSP), Instrumentation Systems

CS / AI: Data Structures & Algorithms, Introduction to AI/ML, DBMS, Computer Architecture & Operating Systems

Mathematics: Probability & Statistics, Linear Algebra, Numerical Methods